Upper limit of docking time for Stikinia and Terrane I: paleomagnetic evidence from the Eocene Ootsa Lake Group, British Columbia

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The Middle Eocene Ootsa Lake Group is exposed in the central portion of the Stikine Terrane, where it was sampled along the shoreline of Tahtsa Reach and Whitesail Reach. The group consists of dominantly subaerial flows, which range in composition from basalt to rhyolite, that unconformably overly the Jurassic Hazelton Group. Detailed alternating-field and thermal stepwise demagnetizations were done on all specimens from the 21 sites collected. The presence of a normal- and reversed-polarity remanence, a positive fold test, and high coercivities and unblocking temperatures indicate that a prefolding primary remanence has been isolated. The mean tilt-corrected direction of $D = 002.2°$, $I = 69.2°$ ($\alpha_{95} = 7.4°$) from 13 sites for which paleohorizontal is well known yields a pole position at 354.6°E, 88.0°N ($A_{95} = 11.5°$), which is statistically indistinguishable from published 50 Ma reference poles for cratonic North America. This evidence demonstrates that the proposed large-scale northward displacement of Stikinia since mid-Cretaceous was completed by at least Middle Eocene time. This result is consistent with other paleomagnetic results from Stikinia, Quesnellia, and the Coast Plutonic Complex indicating that much of the allochthonous Cordillera had assembled and docked with cratonic North America by the Middle Eocene.

Le Groupe d'Ootso Lake de l'Éocène moyen, qui affleure dans la partie centrale du terrane de Stikine, a été échantillonné le long des rives des parties droites Tahtsa et Whitesail. Le groupe, formé principalement de coulées subaériennes de composition de basalte à rhyolite, repose en discordance sur le Groupe d'Hazelton du Jurassique. Une étude détaillée des désaimantations par champ alternatif et par chauffage a porté sur tous les spécimens des 21 sites échantillonnés. La présence de rémanence de polarité normale et inverse, le test du pli positif et les champs coercitifs intenses ainsi que les températures de déblocage indiquent qu'une rémanence primaire acquise avant la phase de plissement a été isolée. La direction moyenne de 13 sites, de paléohorizontalité bien connue, et une fois corrigée sur l'inclinaison, est de $D = 002{,}2°$, $I = 69{,}2°$ ($\alpha_{95} = 7{,}4°$), elle fournit une position du pôle à 354,6°E, 88,0°N ($A_{95} = 11{,}5°$), ne différant pas statistiquement des pôles de la courbe 50 Ma du craton de l'Amérique du Nord. Cette étude démontre que le déplacement du Stikinia sur une grande distance vers le nord depuis le Crétacé moyen s'est terminé avant au moins le temps de l'Éocène moyen. Ces données s'accordent avec les résultats de d'autres études paléomagnétiques du Stikinia, Quesnellia et du Complexe plutonique côtier, indiquant qu'une partie importante de la Cordillère allochtone fut collée et arrimée au craton de l'Amérique du Nord avant l'Éocène moyen.

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Introduction

Numerous paleomagnetic studies in the Cordillera of North America have recognized discordant paleomagnetic directions with respect to the craton. These aberrant directions have been explained by large-scale tectonic rotations and (or) translations of allochthonous terranes within the Cordillera, which have been independently supported to some extent by paleontologic, radiometric, and structural evidence. Much of the paleomagnetic evidence for large-scale northward displacement comes from Cretaceous paleomagnetic data (e.g., Rees *et al.* 1985; Marquis and Globerman 1988), since mid-Tertiary and later Cordilleran data are for the most part concordant.

Currently, the Cretaceous to mid-Tertiary paleomagnetic data base for the Cordillera includes relatively few studies from bedded sequences for which paleomagnetic horizontal is known. Recent paleomagnetic results from bedded sequences include work by Fox and Beck (1985), Symons and Wellings (1989), and Bardoux and Irving (1989). Their results demonstrate that Quesnellia had reached its present latitudinal position relative to North America by Middle Eocene. Although Symons and Wellings (1989) found no evidence of clockwise rotations, Fox and Beck (1985) observed declinations to the north-northeast that they interpreted as indicating 25° of clockwise rotation of their study area. Given the extensional tectonics of that time, this scenario is plausible. Bardoux and Irving (1989) also provided evidence to suggest a clockwise rotation of 28° of their study area if correction for full geologic tilt is applied to all their sites. However, they argued that the Kelowna volcanics magnetizations were acquired over a time spanning deposition and tilting, and therefore correction for full geologic tilt is unwarranted. They interpreted their data using partial tilt corrections for one group of sites, where precision is maximized, and no tilt correction for another group of sites. Their conclusion is consistent with the null hypothesis that, relative to the craton, there have been no large rotations. As pointed out by Kodama (1988), a maximum value of Fisher's (1953) precision parameter (k) at partial unfolding is a necessary, but insufficient, condition to ensure that the magnetization is synfolding in age. Evidence independent of the fold test must be used to substantiate a synfolding age of magnetization. It is evident that in the middle Eocene some areas internal to Quesnellia may have undergone clockwise rotation about vertical axes. Given that the result of Symons and Wellings (1989) is the only unambiguous concordant Middle Eocene pole from bedded volcanics in Quesnellia and Terrane I (Fig. 1) (Monger *et al.* 1982; Monger 1984), further paleomagnetic studies on bedded rocks of this age are needed. Thus, the purpose of this study was to obtain paleomagnetic data from an Eocene volcanic suite with bedding control in order to establish whether large-scale rotations and (or) northward translations were ongoing during this time since the Cretaceous in Stikinia. The Middle Eocene Ootsa Lake Group was chosen for this paleomagnetic study by

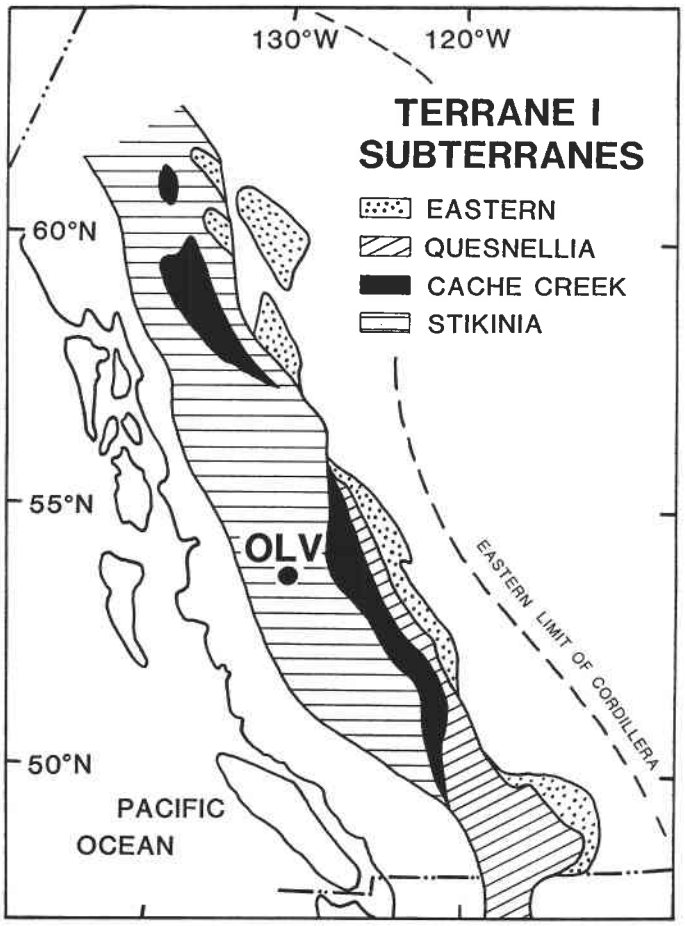


FIG. 1. Location map showing the Ootsa Lake Group study area and its relation to the subterranes of Terrane I from Monger *et al.* (1982). OLV, Ootsa Lake Group volcanics.

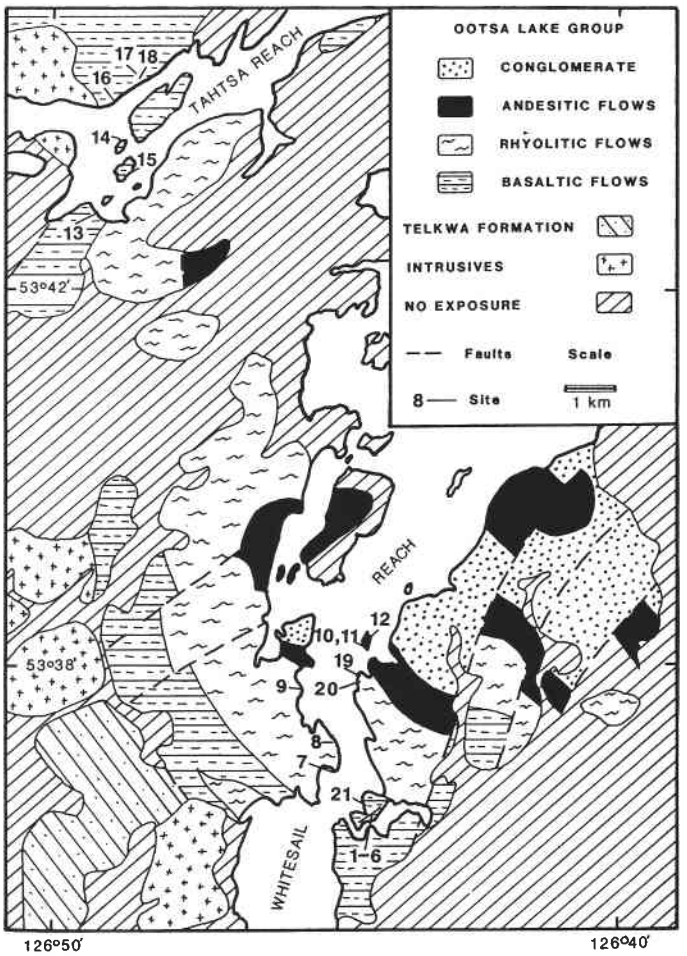


FIG. 2. Geology of the study area after Diakow and Mihalynuk (1987*a*), with site locations indicated.

virtue of its good stratigraphic exposure, age control, lack of post-emplacement intrusives, relatively fresh, unaltered appearance, and location in central Stikina (Fig. 1).

Geology

Much of the following has been discussed by Diakow and Mihalynuk (1987*b*). Stikinia is underlain by mildly deformed and faulted Lower Jurassic to Tertiary volcanic and sedimentary rocks. The Middle Eocene Ootsa Lake Group in the type area around Ootsa Lake comprises rhyolitic flows and less voluminous andesite and basalt, together with associated pyroclastic and sedimentary rocks (Duffell 1959). Recent mapping by Diakow and Mihalynuk (1987*a*) has identified similar Ootsa Lake Group rock types in the Whitesail Range to the west of Ootsa Lake, where subaerial flows, ranging in composition from basalt to rhyolite, and representing over 1000 m of volcanic stratigraphy, dominate. Diakow and Mihalynuk (1987*a*) divided the group into six rock units on the basis of outcrop appearance and lithology. In our study area (Fig. 2) only four of these units crop out: basalts in the lowest unit, overlain by rhyolites, which in turn are overlain by andesites capped by a conglomerate unit.

The Ootsa Lake Group in this area is essentially unmetamorphosed. The youngest intrusives are small Tertiary plugs that cut the Smithers Formation, upon which the Ootsa Lake Group unconformably rests. These plugs are likely intrusive equivalents of the Ootsa Lake Group. Thus, these volcanics were not likely exposed to subsequent thermal events. Localized hydrothermal alteration associated with fractures can be recognized by a cream–white appearance that weathers to rusty fragments. However, we have avoided sampling in these areas. Four K–Ar radiometric dates on biotites from the Ootsa Lake Group are in excellent agreement, ranging between 50.0 ± 1.7 and 49.1 ± 1.7 Ma, and suggest that the group may have erupted over a 1 Ma time span (Diakow and Koyanagi 1988).

Methods

The Ootsa Lake volcanics were sampled at 21 sites (Fig. 2) characterized by shallow- to moderately steep dipping flows of basalt, andesite, and rhyolite. Using a portable drill, we drilled at least five cores at each site and oriented them *in situ* by magnetic and solar compasses. Bedding attitudes were carefully measured as close as possible to each site and recorded. Wherever possible, two specimens were cut from each core. We analyzed the specimens using a Schonstedt DSM-1 spinner magnetometer, Schonstedt GSD-5 tumbling alternating field demagnetizer, and a Schonstedt TSD-1 thermal demagnetizer. After measuring the natural remanent magnetization (NRM), we subjected each specimen to detailed alternating-field (AF) and (or) thermal stepwise demagnetization in up to 14 steps. AF demagnetization was done up to maximum fields of 100 mT, and thermal demagnetization was done up to peak temperatures

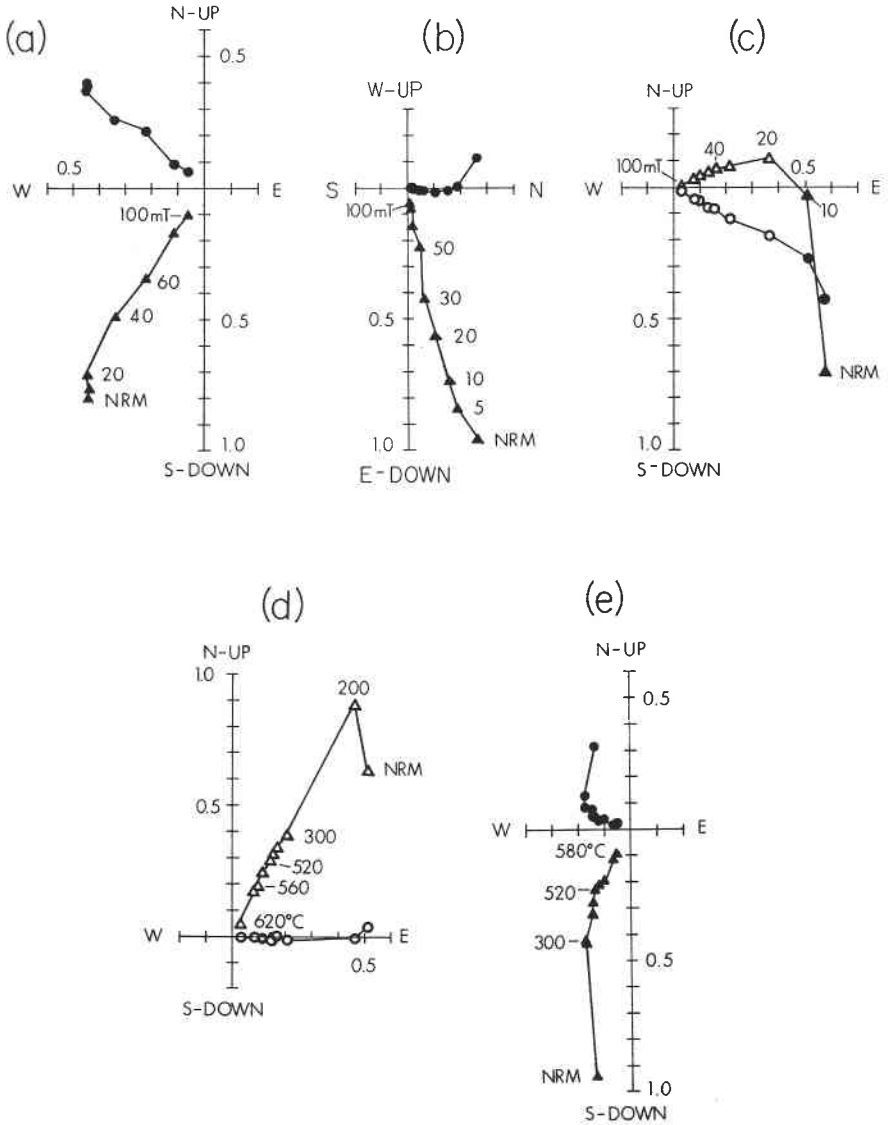


FIG. 3. Orthogonal-vector-decay plots for example specimens on AF and thermal stepwise demagnetization. Circles (triangles) represent vector projections on the horizontal (vertical) planes; solid (open) symbols denote down (up) directions; the AF-step intensities are in milliteslas (mT); thermal-step temperatures are in degrees Celsius (°C). The axial values are given as a ratio of the NMR intensity. The directions are before tilt correction. Specimens are from (*a*) site 12, (*b*) site 3, (*c*) site 14, (*d*) site 5, and (*e*) site 8.

of 700°C. Characteristic remanence directions in each specimen were obtained by a joint analysis of stereographic and orthogonal demagnetization diagrams and least-squares principal-component analysis following Kirschvink (1980). Directions of remanence were accepted with a mean angular deviation (MAD) of $<10°$; most were $<5°$. We calculated the site-mean remanence directions using conventional tiered statistics (Irving 1964; Fisher 1953).

Results

The majority of orthogonal-vector-decay and stereographic plots illustrated data of high quality, demonstrating very stable linear vector decay from all three volcanic rock types (Figs. 3*a*–3*e*). The plots indicated that single stable normal (Figs. 3*a*, 3*b*, and 3*e*) and reversed (Figs. 3*c*, 3*d*) components were isolated after demagnetization removed the "soft" viscous remanence magnetization (VRM) residing in low-coercivity and low-unblocking-temperature domains (Figs. 3*a*–3*e*). The VRM components often differed from one specimen to the next, indicating random directions, some of which were acquired during a storage test in the laboratory. After removal of the VRM components the majority of specimens decayed substantially and linearly towards the origin, with maximum AF's of 100 mT (Figs. 3*a*–3*c*) and (or) thermal unblocking temperatures in both the magnetite and the hematite ranges (Figs. 3*d*, 3*e*). This demonstrates that the same stable remanence is carried by both magnetite and hematite. Stable remanence directions were isolated in each of the 21 sites; their corresponding site means and unit means are shown in Tables 1 and 2.

Fold test

The attitudes of the Ootsa Lake Group volcanics are quite variable, with shallow to moderately steep dips, so a diagnostic fold test (Graham 1949) is possible. A comparison of *in situ* site means (Fig. 4*a*) with tilt-corrected site means (Fig. 4*b*) shows a substantial improvement in the clustering of directions. This much-improved clustering of directions after full tilt correction increases the estimate of k from 11.5 to 32.6, a maximum for

TABLE 1. Site-mean remanence data

Site	Unit	N	D (°)	I (°)	ϕ (°)	ψ (°)	D_C (°)	I_C (°)	R	k	α_{95} (°)
1	Basalt	7	244	72	047	40	034	66	6.959	145	5[a]
2	Basalt	7	248	67	047	40	024	70	6.971	210	4[a]
3	Basalt	6	355	75	045	25	026	54	5.971	171	5[a]
4	Basalt	4	086	−56	055	30	143	−73	3.960	75	10[a]
5	Basalt	4	080	−62	055	30	167	−78	3.986	211	6[a]
6	Basalt	6	106	−73	055	30	200	−67	5.923	65	8[a]
7	Rhyolite	5	296	71	045	30	006	61	4.965	61	11
8	Rhyolite	5	298	64	045	30	354	57	4.965	115	7
9	Rhyolite	5	316	62	065	30	010	58	4.896	39	13[a]
10	Andesite	7	352	69	025	20	007	51	6.619	16	16
11	Andesite	5	324	66	025	20	349	52	4.872	31	14[a]
12	Andesite	8	304	46	048	19	326	47	7.957	162	4
13	Basalt	5	109	−34	105	39	115	−73	4.495	8	29[a]
14	Basalt	9	117	−25	110	45	127	−69	8.905	84	6[a]
15	Basalt	5	132	−37	112	50	196	−73	4.993	593	3[a]
16	Basalt	2	143	−30	130	50	181	−76	1.999	1279	7[a]
17	Basalt	5	159	−37	120	35	191	−58	4.975	163	6
18	Basalt	11	134	−35	108	44	179	−68	10.994	1740	1
19	Rhyolite	7	010	74	025	20	018	55	6.965	171	5
20	Rhyolite	9	012	70	025	14	017	56	8.759	33	9[a]
21	Basalt	8	016	67	038	10	022	57	7.905	73	7

NOTES: N, number of samples; D and I, mean remanence vector's declination and inclination, respectively, relative to present horizontal; ϕ, down-dip azimuth of bedding; ψ, dip angle from horizontal; D_C and I_C, same as D and I but after tilt correction relative to bedding; R, resultant vector for the summation; k and α_{95}, precision parameter and radius of the circle of 95% confidence, respectively (Fisher 1953).

[a]Sites with *in situ* measured attitudes.

TABLE 2. Uncorrected and corrected unit means and their paleopoles

C or U mean	N	D (°)	I (°)	k	α_{95} (°)	d_p (°)	d_m (°)	Long.	Lat.
U	21	309.3	61.2	11.5	9.8	11.6	15.0	214.7°W	55.1°N
C	21	002.5	64.8	32.6	5.7	7.4	9.2	39.3°E	82.6°N
U	13	298.8	60.3	10.7	13.3	15.3	20.2	208.2°W	48.2°N
C	13	002.8	69.1	32.3	7.4	10.7	12.6	5.4°W	88.0°N

NOTES: C, tilt-corrected unit mean; U, uncorrected unit mean; N, number of site means; D, I, k, and α_{95} as in Table 1; d_p, radius of the ellipse of 95% confidence at the pole along the site–pole great circle; d_m, radius of this ellipse perpendicular to the site-pole great circle.

the group mean. The variance-ratio test (Watson 1956) indicates a positive fold test at the 99% confidence level (McElhinny 1964). McFadden and Jones (1981) suggested that generally this test is too rigorous. Thus, the fold test asserts with at least 99% confidence that the remanence is prefolding in origin.

Magnetostratigraphy

Nine of the 21 sites studied (sites 4–6 and 13–18, Fig. 2) are of reversed polarity, and they all occur in the basalt flows of the lowest stratigraphic unit sampled. Basalt sites 1–6 on Whitesail Reach are stratigraphically ordered in successive flows downward through about 200 m of section. Sites 1–3 in the upper half are normally polarized, whereas 4–6 in the lower half are reversely polarized and likely record the same reversed chron as found in basalt sites 13–18 on Tahtsa Reach. Previously, sites 13–18 could only be roughly correlated on the basis of their basaltic composition. However, given that a reversal of the Earth's magnetic field only takes a few thousand years (Hospers 1954; Hoffman 1988), this stratigraphically confined reversed chron provides an accurate marker horizon for more precise stratigraphic correlation. Since the rhyolite and andesite units are stratigraphically above the basalt unit and are characterized by only normal polarity, this must represent the same normal-polarity chron observed in the upper basalt flows. This follows, given that the Ootsa Lake Group volcanics erupted over about a 1 Ma time span, with volcanic activity ending around 49.1 ± 1.7 Ma (Diakow and Koyanagi 1988), and polarity chron 21, spanning 47.46–49.91 Ma (Harland *et al.* 1982), is the only reasonable time-correlative chron. Polarity chron 20 is too young, spanning 44.31–47.46 Ma, and polarity chron 22 is too old, spanning 49.91–51.39 Ma (Harland *et al.* 1982). The two polarity chrons are assigned to the 21 reversed and 21 normal chrons (Harland *et al.* 1982) based on their excellent fit with the radiometric ages for the Ootsa Lake Group.

The isolation of a magnetization with both polarities is very convincing evidence that the remanence is primary, and if both polarities are combined, secular variation should be averaged out. The mean normal- and reversed-polarity remanence are roughly antipodal after tilt correcting each site and are separated by only 14° of arc.

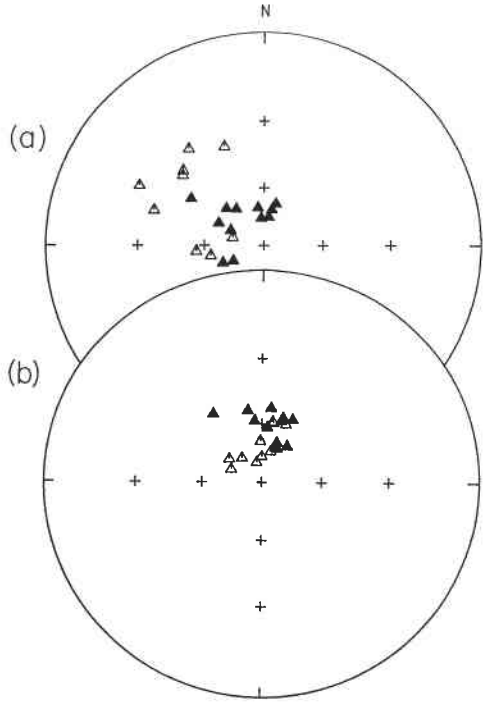


FIG. 4. Equal-area stereographic projection showing (*a*) the site-mean directions relative to present horizontal; and (*b*) the same, but tilt corrected relative to bedding. All triangles are downwardly directed vectors; however, those with open symbols have been converted from their antipodal reversed position.

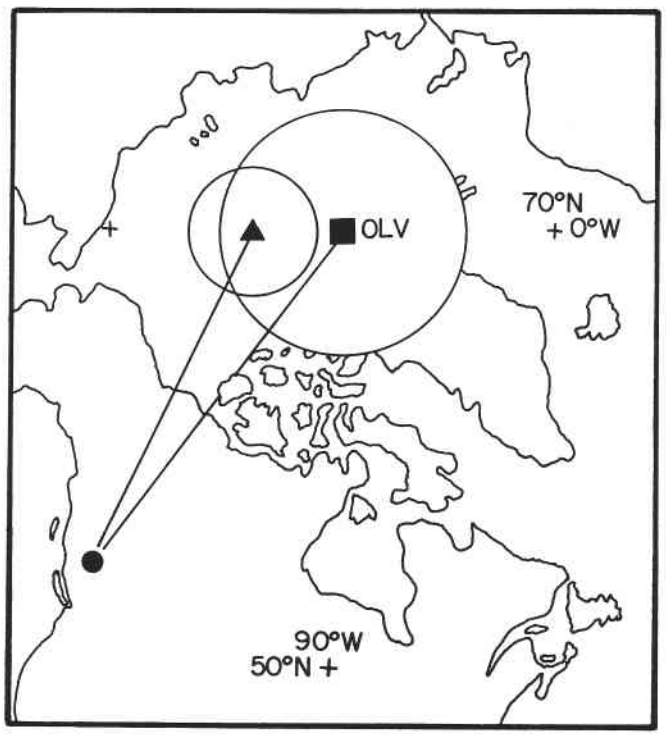


FIG. 5. Pole position for the Ootsa Lake Group volcanics (OLV) (corrected 13 sites, Table 2) shown by a square with its circle of 95% confidence. The study area is shown by the solid circle. The 50 Ma reference pole for the North American craton is shown by a triangle with its circle of 95% confidence (Bardoux and Irving 1989).

Discussion

Given a single stable remanence direction spanning two polarity chrons and a positive fold test at the 99% confidence level, we interpret this as a primary magnetization acquired upon cooling of the lava flows. The relatively small directional difference between normal and reversed polarities may suggest that secular variation has not been adequately averaged out within a single polarity. However, when data from the two polarities are combined, giving a k value of 32 with 95% confidence limits of $17 \leq 32 \leq 53$ (Cox 1969), secular variation appears to have been averaged out, since the expected k value of 20 (McFadden and McElhinny 1984) falls within this range at 95% confidence.

Since we are concerned with the tectonic implications of these results, the importance of accurately knowing the paleomagnetic horizontal of the sampling sites cannot be overemphasized. Equally important is accurately knowing the corresponding reference pole for cratonic North America.

There seem to be two philosophical approaches amongst the paleomagnetic community regarding the interpretation of paleomagnetic horizontal in bedded rocks. One approach prefers to apply well-understood regional bedding attitudes as tilt corrections for paleomagnetic sampling sites in an area. Another approach prefers accurate bedding measurements made *in situ* at each individual sampling site. Together, both attempt to accurately establish the most representative tilt corrections for prefolding and (or) synfolding magnetizations. In this study, because of quite variable bedding strikes and dips from one sampling locality to the next, the best tilt corrections should be obtained from accurately measured *in situ* bedding attitudes at each site. Unfortunately, this could only be done at 13 of the 21 sites collected for this study, so we calculated a second unit mean (Table 2) using only those sites where bedding attitudes could be directly measured. Importantly, this second unit mean from 13 sites combines six normal-polarity and seven reversed-polarity sites and minimizes the influence of bedding-attitudes errors, which can introduce large errors into the tectonic analysis of the data. It is possible that primary dips could be incorporated into these tilt corrections; however, a significant primary dip will manifest itself by high dispersion both before and after tilt correction, which is not the case here.

Tectonic analysis

Since popular North American reference poles calculated by a moving average-window technique may smooth out polar-wander details, it is more accurate to look at the data base specifically applicable to the Middle Eocene and narrow the time-averaging window within the tightest possible limits. This has been done by Bardoux and Irving (1989), who provided a reference pole (180°E, 83°N, A_{95} = 6.0°) from three well-dated studies that constrain the age of the pole to between 47 and 53 Ma.

The above reference paleopole of Bardoux and Irving (1989) and that for the Ootsa Lake Group (corrected 13 sites, Table 2) are in good agreement (Fig. 5). Carrying out the tectonic analysis outlined by Yole and Irving (1980), incorporating the error analysis of Demarest (1983), demonstrates that the Ootsa Lake Group paleopole is statistically indistinguishable from the cratonic reference pole for North America. Relative paleolatitude displacement is statistically nonsignificant at 4.8° ± 10.2° to the north, and relative rotation is also statistically nonsignificant at 13.2° ± 17.5° clockwise. Note that this result does not

change significantly if the 50 Ma reference poles from Irving and Irving (1982), Harrison and Lindh (1982), or Diehl *et al.* (1983) are used. The statistically nonsignificant magnitudes of northward displacement and clockwise rotation are arguably beyond the attainable precision of any single paleomagnetic study and are therefore, for all intents and purposes, equal to zero. This analysis indicates that Stikinia was in much the same latitudinal position relative to the North American craton during the Middle Eocene as it is at present.

Geotectonics

Stikinia is thought to have been assembled with several other subterranes in the Middle Triassic to form the allochthonous microplate Terrane I (Monger *et al.* 1982). As mentioned earlier, recent paleomagnetic studies on Middle Eocene volcanics from one of these subterranes, Quesnellia (Fox and Beck 1985; Symons and Wellings 1989; Bardoux and Irving 1989), have demonstrated that Quesnellia, which is inboard of Stikinia in Terrane I (Fig. 1), was in much the same latitudinal position in the Middle Eocene as it is now. It is significant that the Ootsa Lake Group, located in Stikinia, has not been rotated clockwise, unlike some areas in Quesnellia that may have been. This is likely due to its more central location in Terrane I, where there was greater stability. This is the first Middle Eocene paleomagnetic result from Stikinia. A previous study on the younger Late Miocene Cariboo plateau lavas and Cariboo gabbro plugs that overlie and intrude the Stikine, Quesnel, and Cache Creek subterranes also yielded concordant results (Symons 1969*a*, 1969*b*), consistent with the result from this study. Further middle Tertiary paleomagnetic data from intrusions of the Coast Plutonic Complex immediately west of Terrane I are also concordant, with the exception of a low-accuracy discordant result from the Mount Barr Complex (Symons 1973*b*). These concordant results include those from the Chilliwack batholith (Beck *et al.* 1982); the Hope Complex (Symons 1973*b*); the Grotto and Snoqualmie batholiths (Beske *et al.* 1973); the Spuzzum–Porteau plutons overprint (Irving *et al.* 1985); the Brown basalt dikes (Symons 1968); and the north-northwest-trending plutons of the Hawkesbury warp (Symons 1977). Middle Tertiary studies from rocks farther west in the Coast Plutonic Complex and Wrangellia do show divergences (e.g., Symons 1973*a*; Hicken and Irving 1977). However, in the main body of the Canadian Cordillera, paleopoles from volcanics and intrusives of mid-Tertiary age and younger are consistent with our result from Stikinia. This is strong evidence that stabilization and docking of Terrane I and much of the Canadian Cordillera with North America took place by at least the Middle Eocene.

Conclusions

The Ootsa Lake Group is characterized by a stable, two-polarity, prefolding, primary remanence that was acquired upon cooling when the lavas were extruded around 50 Ma. The polarity chrons correspond to 21 normal and 21 reversed chrons (Harland *et al.* 1982) and provide a useful stratigraphic marker horizon for the Ootsa Lake Group. The Ootsa Lake Group paleopole is statistically indistinguishable from several current estimates of the comparable cratonic reference pole. This tectonic analysis indicates that Stikinia was in much the same latitudinal position relative to the North American craton during the Middle Eocene as it is at present. This demonstrates that the proposed large-scale northward displacement of Stikinia since the mid-Cretaceous was completed by at least Middle Eocene time. This docking time is consistent with concordant paleomagnetic results from Stikinia, Quesnellia, and the Coast Plutonic Complex that also indicate that much of the allochthonous Cordillera had assembled and docked with North America by at least Middle Eocene.

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